

TESTING THE LINEAR REGRESSION MODEL - RESIDUALS

On establishing a linear relationship between two variables on the basis of the correlation coefficient in conjunction with the scatterplot of the data, a line of best fit is fitted to the data to enable predictions to be made. This line of best fit is obtained by the "method of least squares" and we call this line of best fit the linear regression model.

This linear regression model may or may not be suitable for prediction purposes as a high correlation coefficient between the two variables does not necessarily imply that the linear regression model is the most suitable for prediction purposes.

In order to determine the suitability of a linear regression model for prediction purposes we need to examine the **residuals** by constructing a **residual plot**.

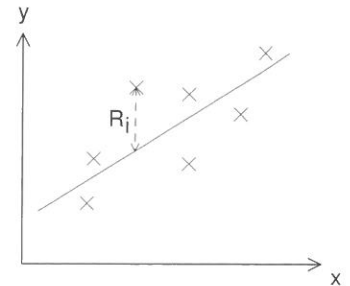
The diagram on the right shows the regression line and a residual R_i .

A **residual** is the difference between an observed data point and a fitted or predicted value. It is the distance of a point from a curve, and is positive if above the curve and negative if below the curve. The residual R_i is calculated as follows:

That is $R_i = \text{observed value} - \text{fitted value}$.

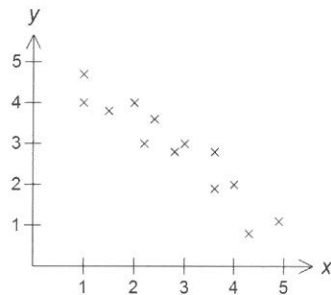
$$= y_i - \hat{y}_i$$

where $\hat{y}_i = ax_i + b$ is the value predicted by the linear regression model.

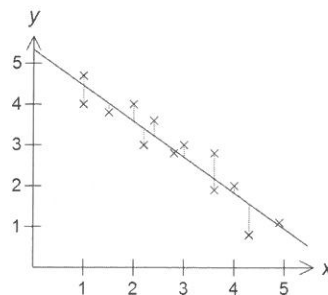


Consider the following bivariate situations:

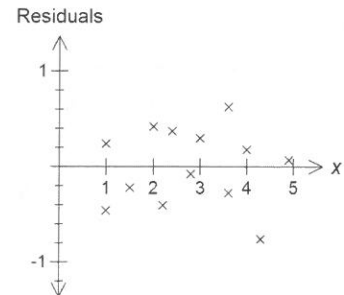
Situation 1



Graph A scatterplot



Graph B regression line



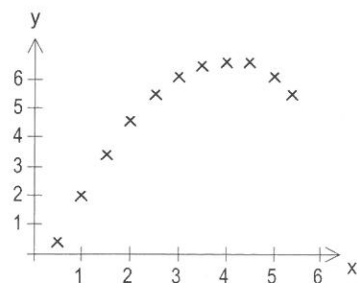
Graph C residual plot

Graph A shows the distribution of bivariate data and the relationship between the variables x and y appears to be a strong negative linear relationship. The correlation coefficient $r_{xy} = -0.94$ supports the scatterplot.

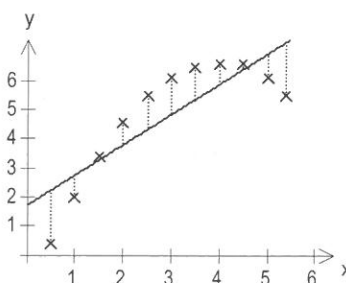
Graph B shows the fitted regression line $\hat{y} = -0.88x + 5.34$ and the residuals, showing that the residuals are randomly scattered and there is no discernible pattern. Hence the linear regression model suits the given data and is appropriate model for regression (making predictions).

Graph C is a residual plot with the x -axis the same as that for the scatterplot and the y -axis showing the possible magnitudes of the residuals. The residual plot gives the same information as graph B but it is much easier to see the random distribution of the residual values.

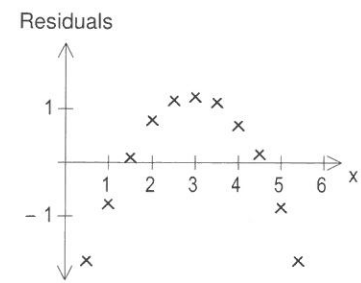
Situation 2



Graph A scatterplot



Graph B regression line



Graph C residual plot

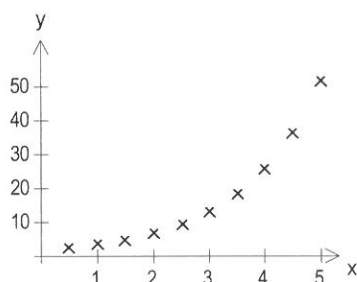
Graph A shows the distribution of bivariate data and the relationship between the variables x and y is not linear and appears to be a strong quadratic relationship. The correlation coefficient for this scatterplot is given as $r_{xy} = 0.83$ indicating that there is a strong positive relationship between x and y .

The correlation coefficient does not support the scatterplot and it can be clearly seen that a high correlation coefficient does not necessarily imply that the relationship under consideration is linear. Thus consideration of only the correlation coefficient is not sufficient to inform us about the suitability of linear regression.

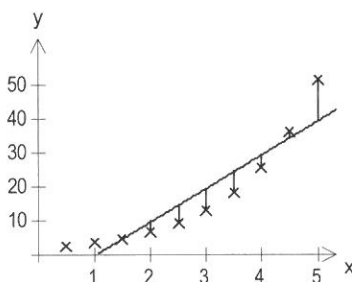
Graph B shows the fitted regression line $\hat{y}=1.04x+1.72$ and the residuals, showing that the residuals are not randomly scattered above and below the curve and there is a discernible pattern. For low values of x the residuals are below the regression line that is negative, for middle values of x the residuals are above the regression line that is positive and for high values of x the residuals are negative. Hence using the linear regression model $\hat{y}=1.04x+1.72$ would not be appropriate for regression as the relationship between x and y is clearly not linear. Regression for this data set must be done using a different regression model as linear regression is not appropriate. Perhaps quadratic regression may be appropriate in this case.

Graph C the residual plot gives the same information as graph B but it is much easier to see the non-random nature of the distribution of the residual values and the magnitude of the residuals.

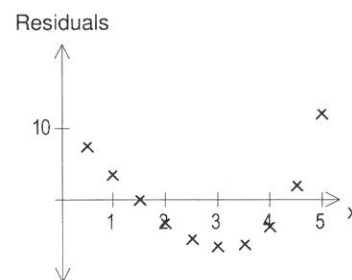
Situation 3



Graph A scatterplot



Graph B regression line



Graph C residual plot

Graph A shows the distribution of bivariate data and the relationship between the variables x and y is not linear and appears to be a strong exponential relationship. The correlation coefficient for this scatterplot is given as $r_{xy}=0.92$ indicating that there is a strong positive relationship between x and y .

The correlation coefficient does not support the scatterplot and it can be clearly seen that a high correlation coefficient does not necessarily imply that the relationship under consideration is linear. Thus consideration of only the correlation coefficient is not sufficient to inform us about the suitability of linear regression.

Graph B shows the fitted regression line $\hat{y}=9.89x-10.26$ and the residuals, showing that the residuals are not randomly scattered above and below the curve and there is a discernible pattern. For low values of x the residuals are positive, for middle values of x the residuals are negative and for high values of x the residuals are positive. Hence using the linear regression model $\hat{y}=9.89x-10.26$ would not be appropriate for regression as the relationship between x and y is clearly not linear. Regression for this data set must be done using a different regression model as linear regression is not appropriate, perhaps exponential regression may be appropriate in this case.

Graph C the residual plot gives the same information as graph B but it is much easier to see the non-random nature of the distribution of the residual values and the magnitude of the residuals.

In summing up:

- The correlation coefficient on its own is a very poor indicator in the validation of a linear regression model and on its own cannot be used to justify a linear relation between two variables. If used with an accompanying scatter diagram its use is strengthened to some degree but it is still not an acceptable justification as the relationship may not be linear.
- A residual plot will allow us to determine the validity of the linear regression model as a predictor. As residuals are like random errors the residual plot should be a random scatter of points in the plane, any evidence of a non-random distribution (or pattern) in the residual plot would indicate that our linear regression model is not the curve of best fit. If a pattern or a non-random distribution is evident in the residual plot then another regression model must be used for prediction purposes, for example, a least squares parabola, that is a quadratic regression model may be appropriate.
- The magnitude of the residuals gives us an indication of the strength of the linear relationship between the two variables. The smaller the magnitude of the residuals the higher the correlation between the two variables resulting in an increase in the reliability of any predictions that are made.
- Any outliers in the residual plot have to be checked out and if found to be real and not as a result of misreporting or other error then either the outlier or the linear regression model must be discarded.
- When making predictions the reliability of the prediction may be questionable if the number of data points being used is small.

Example 8

The Test 1 and Test 2 results of the Unit 3 Applications course for a class are shown below. Consider the suitability of a linear model for predicting a Test 2 score for a given Test 1 score.

Student	1	2	3	4	5	6	7	8	9	10	11	12
Test 1(x)	67	50	75	70	65	85	91	57	33	66	93	79
Test 2(y)	82	35	86	81	76	88	85	60	50	68	86	82

From the calculator we obtain $a = 0.8171986$

$b = 16.658999$

$r = 0.8397605$

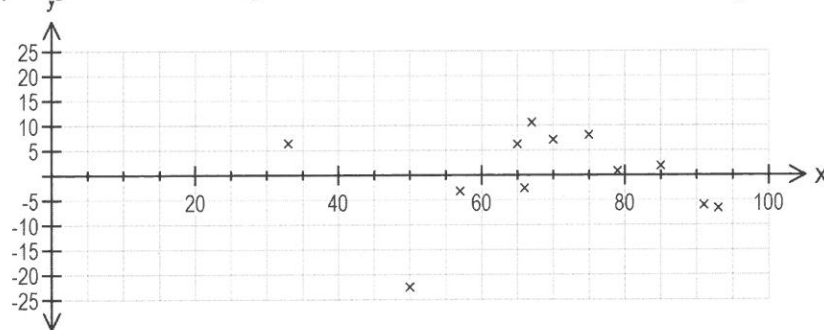
Hence the linear regression model for y on x is given by $\hat{y} = 0.8171986x + 16.658999$

To calculate the residuals we make use of the following table:

Student	1	2	3	4	5	6	7	8	9	10	11	12
Test 1(x)	67	50	75	70	65	85	91	57	33	66	93	79
Test 2(y)	82	35	86	81	76	88	85	60	50	68	86	82
$\hat{y}$	71.4	57.5	77.9	73.9	69.8	86.1	91.0	63.2	43.6	70.6	92.7	81.2
$y - \hat{y}$	10.6	-22.5	8.1	7.1	6.2	1.9	-6.0	-3.2	6.4	-2.6	-6.7	0.8

Where $\hat{y}$ are the predicted values given by $\hat{y} = 0.81719856x + 16.6589991$ and $y - \hat{y}$ are the residuals.

Graphing the residuals against the x values we obtain the following residual plot.



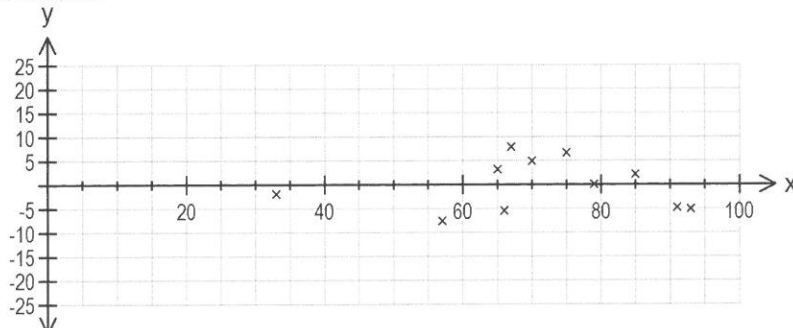
The residual plot does not suggest any definite pattern, thus the linear regression model given by $\hat{y} = 0.81719856x + 16.6589991$ appears to be suitable for this bivariate distribution.

However closer examination of the residuals suggests that the residual for the student numbered 2 with ordered pair (50, 35) is comparatively large and hence should be checked out.

In this case the point is real, that is no errors have been made, hence we would use this model for prediction of a test 2 score for a given test 1 score or consider a different regression model.

However if it was found that student numbered 2 had been absent for an extended period of time prior to test 2 due to illness then in this case there is justification for cropping the data set as there is evidence of a characteristic which is not present in the rest of the distribution.

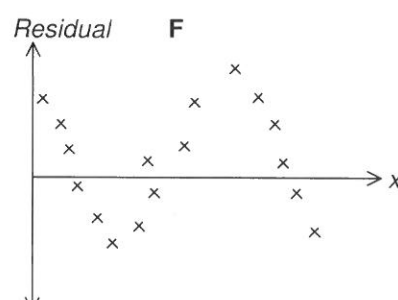
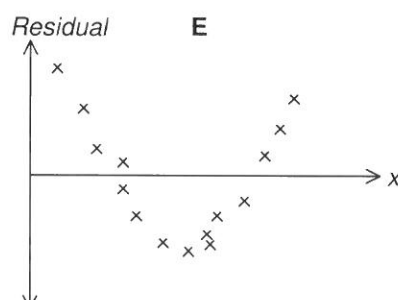
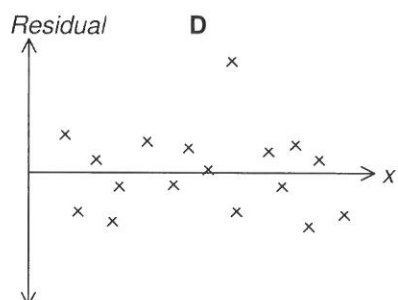
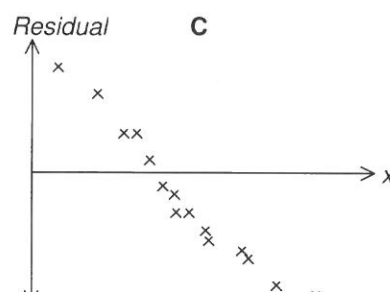
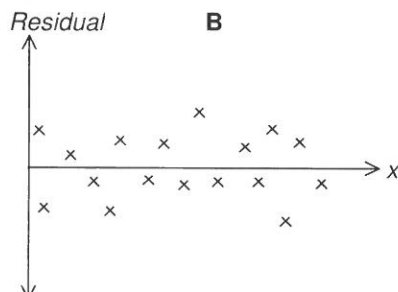
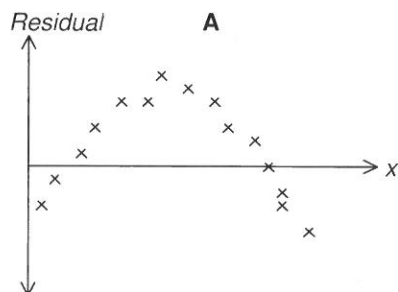
On removal of student 2 the revised linear regression model is given by $y = 0.6528839x + 30.372512$ and the resulting residual plot, shown below does not suggest any definite pattern indicating that the linear regression model is suitable for prediction purposes. Furthermore, with the removal of the outlier, student numbered 2, the value of the correlation coefficient has increased from 0.8397605 to 0.90069847 supporting the suitability of the linear model.



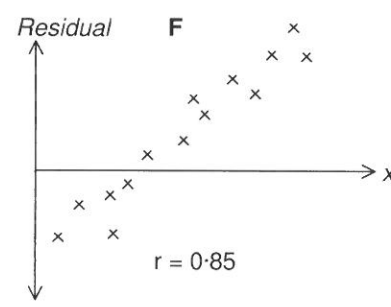
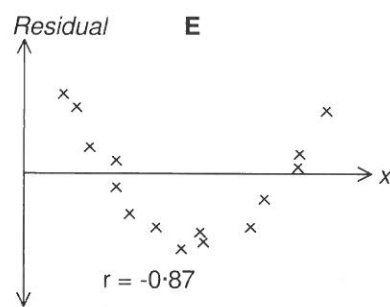
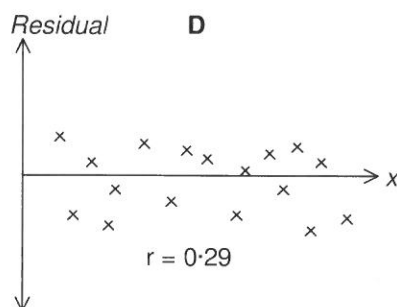
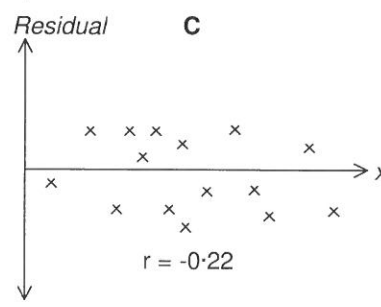
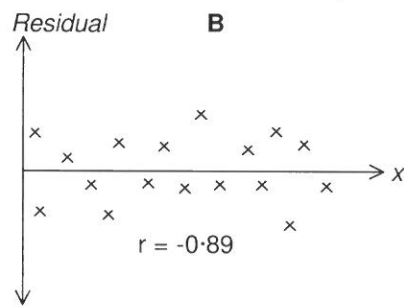
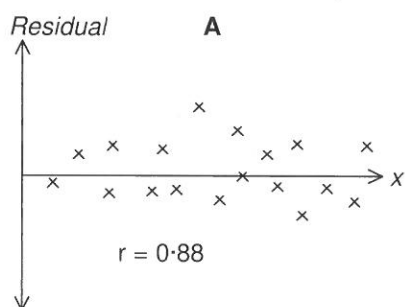
Note that the magnitude of the residuals in the cropped residual plot are much smaller than that of the original data set further supporting the appropriateness of the cropped linear model.

EXERCISE 3E

1. Consider the following residual plots and comment on the suitability of the linear regression model associated with the residual plot.



2. Consider the following residual plots and the associated correlation coefficient. For each case state whether the linear regression model would be suitable for prediction purposes.



4. Consider the following bivariate distribution.

x	1	1	2	4	4	5	5	6	7	8	8	9	9
y	1	3	3	3	4	3	4	5	5	4	6	5	6

- Find the coefficient of correlation correct to four decimal places and comment on the relationship between the variables x and y .
- Determine the linear regression line of y on x .
- Examine the residual plot and comment on the suitability of the linear regression model.
- Predict a value for y when $x = 10$ and comment on the reliability of your prediction.
- Predict a value for y when $x = 20$ and comment on the reliability of your prediction.

5. Consider the following data.

x	1	2	3	7	12	18
y	9	7	6	5	3	1

- Examine the scatter plot of the given data set and comment on the relationship between x and y .
- Find r_{xy} and comment on the relationship between x and y .
- Find the least squares regression line of y on x .
- Construct a residual plot and comment on the suitability of using the least squares regression line of y on x for predicting the value of y for $x = 9$.

6. Consider the following data.

x	1	2	3	4	5	6	7	8	9	10	11	12
y	360	700	1100	1465	2000	2222	2809	3355	4167	4833	5532	6201

- Examine the scatter plot of the given data set and comment on the relationship between x and y .
- Find r_{xy} and comment on the linear relationship between x and y .
- Find the least squares regression line of y on x .
- Construct a residual plot and comment on the suitability of using the least squares regression line of y on x for predicting the value of y for $x = 4.6$.

7. Consider the following data.

x	3	4	5	6	7	8	9	10	11	12
y	150	62	32	16	8	4	2	1	0.5	0.2

- (a) Examine the scatter plot of the given data set and comment on the relationship between x and y .
- (b) Find r_{xy} and comment on the linear relationship between x and y .
- (c) Find the least squares regression line of y on x .
- (d) Construct a residual plot and comment on the suitability of using the least squares regression line of y on x for predicting the value of y for $x = 6.5$.

8. A class of music students sat for their Theory (t) and Practical (p) examinations, their results expressed as a percentage have been listed in the spreadsheet below.

	A	B	C	D	E	F	G	H
1	Student	Theory (t)	Practical (p)	$\hat{p}$	$p - \hat{p}$	$\hat{t}$	$t - \hat{t}$	
2	1	58	56					
3	2	61	63					
4	3	65	66					
5	4	69	70					
6	5	74	71					
7	6	77	75					
8	7	80	83					
9	8	87	80					
10	9	91	91					
11	10	93	87					
12								

- (a) Using a graphics calculator or a suitable computer programme examine the scatter diagram of the above data and then comment on the relationship between the practical music marks and the theory music marks.

- (b) Find a linear regression model for prediction of practical marks given a music theory mark.

On the spreadsheet $\hat{t}$ represents the predicted theory mark and $\hat{p}$ represents the predicted practical mark.

- (c) Give the following spreadsheet entries for each of the following cells.

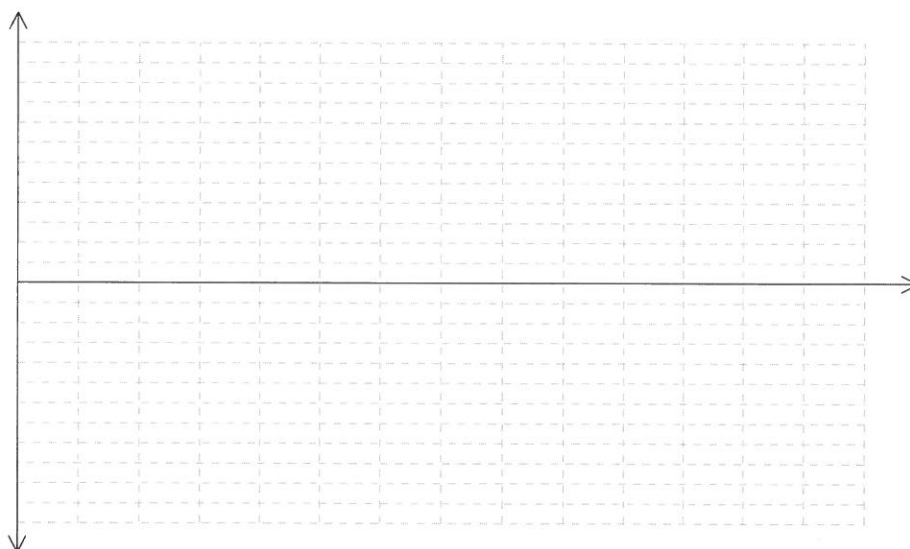
(i) Cell D2

(ii) Cell E2

(iii) Cell F2

(ii) Cell G2

- (d) Complete the spreadsheet by making entries in all of the empty cells.
- (e) What information is contained in column E of the spreadsheet?
- (f) What use can be made of the column E entries.
- (g) Construct a residual plot on the given axes and comment on the suitability of using the least squares regression line found in (b) for making predictions.

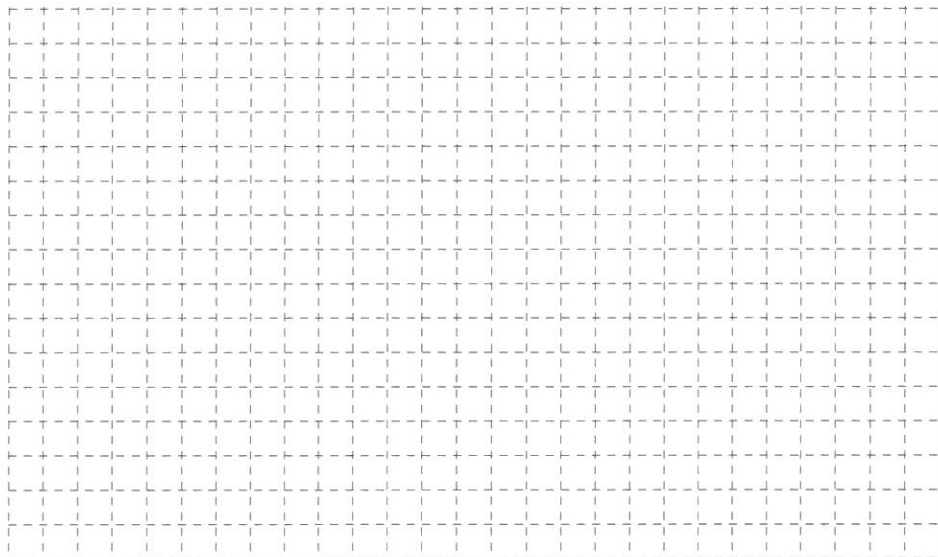


- (h) A student scored 90 marks in her theory examination but due to a hand injury she could not sit the practical examination. What practical mark should she be awarded on the basis of her performance in the theory examination?
- (i) What theory mark should a student be awarded if the student received a practical mark of 87?
- (j) Justify the suitability of your regression model for the prediction of the theory mark.

9. A study of teenagers revealed that the linear correlation coefficient between level of obesity and number of hours spent using a computer was 0.97.
- According to the result of this study what percentage of the variation in the level of obesity could be accounted for by the variation in the number of hours spent using a computer.
 - Does this study suggest that by increasing the number of hours a teenager spends using a computer then the teenagers level of obesity will increase. Justify your response.

10. Hot food is put into a refrigerator. The refrigerator is kept at a constant temperature of 3°C . The temperature of the food is checked every 10 minutes and it is recorded to the nearest degree.

Time elapsed (t mins)	10	20	30	40	50	60	70	80	90	100
Temperature ($^{\circ}\text{C}$)	48	38	30	4	18	14	9	5	4	3



- On the grid above draw a scatter plot representing the information in the table.
- Describe the relationship shown in the scatter plot.
- Calculate and interpret the value of the correlation coefficient between elapsed time and temperature.

From your graph it should be obvious that one data point does not fit the trend of the data.

- Identify this data point.

It was found that the data point in part (d) was a recording error and it should have been recorded as data point (40, 24).

- Without performing any calculations what will be the effect on the value of the correlation coefficient if the recording error is corrected?

- (f) Correct the recording error and then calculate the value of the correlation coefficient.
- (g) Find the equation of the least squares regression line for predicting the temperature of the food for elapsed time. Give the parameters of your equation rounded to 3 decimal places.
- (h) Identify and interpret the y-intercept of your regression line in context.
- (i) Identify and interpret the gradient of your regression line in context.
- (j) Calculate the value of the coefficient of determination and use it together with the correlation coefficient determined in part (f) to determine if the linear regression model in part (g) is appropriate for making predictions.
- (k) Use your regression model to predict the temperature of the food, to one decimal place, after being in the refrigerator for 1 hour.
- (l) Calculate the residual value for elapsed time of one hour.
- (m) On the given axes construct a residual plot for the corrected data set.



- (n) Does your residual plot support your answer to part (j) above? Justify your response.

11. The owner of a free-range egg farm noted that there appeared to be a relationship between the number of eggs laid and the average daily temperature. The table below gives the weekly average number of eggs laid (n) by each hen and average weekly temperature (t) in degrees Celsius over a ten-week period.

Week	1	2	3	4	5	6	7	8	9	10
n	5.9	6.6	5.7	6.1	7	7.5	8	7.6	8.1	7.9
t	19.4	19.5	19.3	19.5	19.6	19.8	19.9	19.9	19.8	19.8

- Identify the explanatory variable and the response variable.
- Find the value of r_{nt} . What does the value of r_{nt} indicate?
- Calculate the line of regression that can be used to predict the average laying rate from the average weekly temperature.
- State the gradient of the regression line and explain what it means.
- State the vertical axis intercept of the regression line and state what it means.

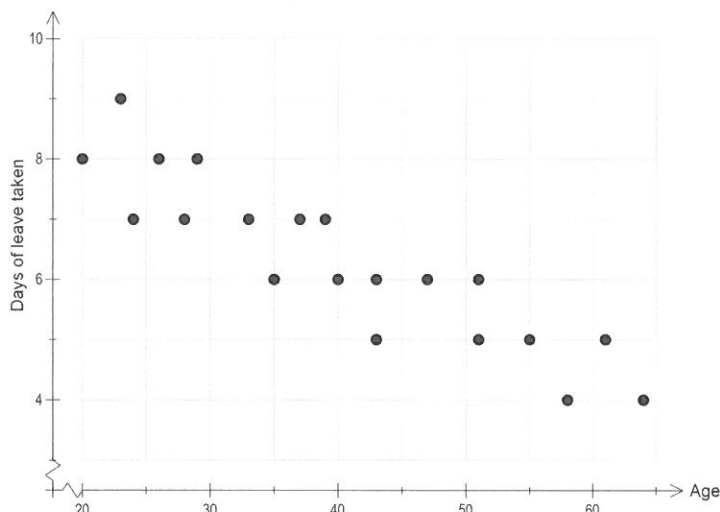
In order to check on the reliability of the regression model it is necessary to examine the residual plot for the model in question.

- On the grid below draw a residual plot for this model and comment on how it is used to validate the reliability of prediction.



- Using your prediction model determine the average number of eggs that can be expected if the average weekly temperature is 20°C .
- Discuss the reliability of your prediction.

12. In a study between the age, in years, of all employees in a small company, and the number of days sick leave taken last year by these employees resulted in the graph below.



The correlation coefficient for this data is -0.9060 .

The equation of the least squares regression line for predicting the number of days sick leave is:

$$\text{Number of days leave} = 10.093 - 0.094 \times \text{age}$$

- Graph the regression line on the scatter plot above showing two key points used in drawing the line.
- State the gradient of the regression line and explain what it means.
- State the vertical axis intercept of the regression line and state what it means.
- Explain the relationship between the sign of the correlation coefficient and gradient of the least squares regression line.
- Calculate the coefficient of determination and state what it tells you.
- Predict how many days a fifty-five year old employee might be expected to be absent.
- Comment on the reliability of your prediction. Give reasons.
- Find the residual when the age of an employee is 55 years.
- Explain what your answer to part (f) means.